

VP150 Final RC

Energy and Momentum

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Work, Power, and Energy

Equations

$$\text{Work: } W_{AB} = \int_{\Gamma_{AB}} \vec{F} \circ d\vec{r}$$

$$\text{Average Power: } P_{av} = \frac{W}{\Delta t} = \vec{F} \circ \vec{v}_{av}$$

$$\text{Instantaneous Power: } P(t) = \frac{dW(t)}{dt} = \vec{F} \circ \vec{v}$$

$$\text{Kinetic Energy: } K = \frac{1}{2}mv^2$$

$$\text{Work-Kinetic Energy Theorem: } W = \Delta K$$

Momentum

Definition

$$\vec{p} = m\vec{v} (\text{Unit : kg} \cdot \text{m/s})$$

Derived Formulas



$$\vec{F} = \frac{d\vec{p}}{dt} \text{ (Follows from Newton's II law)}$$



$$p_2 - p_1 = \int_{t_1}^{t_2} F dt \text{ (Momentum-Impulse Theorem)}$$

Conservation of Momentum

Formula

$$\vec{p} = \sum \vec{p}_i = \text{const}$$

Explanation

- No external force on the system lead to $\Rightarrow$ No change in the momentum of the system
- The momentum of part of the system might change.
- Remember that momentum is a vector, which can be decomposed.

Collision

Classification

Elastic collision: $K_{final} = K_{initial}$

Inelastic collision: $K_{final} < K_{initial}$

Completely inelastic collision: stick to each other

Superelastic collision: $K_{final} > K_{initial}$

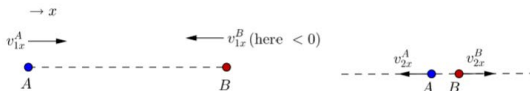
How to solve problems

- Elastic collision: Conservation of Kinetic Energy and Momentum
- Completely inelastic collision: Conservation of Momentum

Elastic Collision Case

How to solve

General 1D case (head-on collision) → before/after the collision the velocities of both particles are colinear



- ① Momentum conserved

$$m_A v_{1x}^A + m_B v_{1x}^B = m_A v_{2x}^A + m_B v_{2x}^B$$

- ② Energy conserved

$$\frac{1}{2} m_A (v_{1x}^A)^2 + \frac{1}{2} m_B (v_{1x}^B)^2 = \frac{1}{2} m_A (v_{2x}^A)^2 + \frac{1}{2} m_B (v_{2x}^B)^2$$

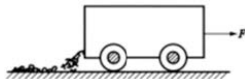
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Exercise 1

As is show in the figure, a person pulls a car filled with coal in a constant force F . Since there is a leak hole in the bottom of the car, the coal is leaking out of the car with a constant speed of $q(\text{kg/s})$. Assume that the car is originally stationary, and the mass is m_0 (together with the coal). The person begins pulling the car at $t = 0$. The coal begin leaking at the same time. Try to find the speed of the car at time t .



Solution 1

Since according to the Momentum-Impulse Theorem,

$$Fdt = dp = (m - qdt)(v + dv) + qdt \cdot v - mv \approx mdv$$

Therefore,

$$F = m \frac{dv}{dt} = (m_0 - qt) \frac{dv}{dt}$$

Therefore,

$$\frac{dt}{m_0 - qt} = \frac{dv}{F}$$

By integration,

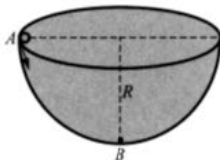
$$\frac{1}{q} \ln\left(\frac{m_0}{m_0 - qt}\right) = \frac{v}{F}$$

Finally,

$$v = \frac{F}{q} \ln\left(\frac{m_0}{m_0 - qt}\right)$$

Exercise 2

As shown in the picture, a particle of mass m is in a hemispherical container of radius R . It is sliding from a point A on the edge from rest, and when it reaches the lowest point B, the pressure on the container is N . Calculate the work done by the friction force in the process of the mass sliding from the point A to point B.



Solution 2

In the lowest point B,

$$m \frac{v^2}{R} = N - mg$$

From A to B, apply the work-kinetic energy theorem,

$$\frac{1}{2}mv^2 - 0 = mgR + W_f$$

Combining these two equations, we can calculate

$$W_f = -\frac{3}{2}mgR + \frac{1}{2}NR$$